Situation Calculus

Reasoning about action and agent programming

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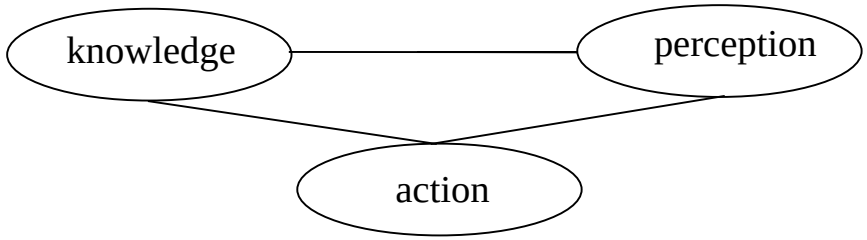
Outline

Disclaimer: some slides are based on were "stolen" from Hector Levesque, Giuseppe de Giacomo, and Ryan Kelly, and others. :-)

Outline

Cognitive Robotics

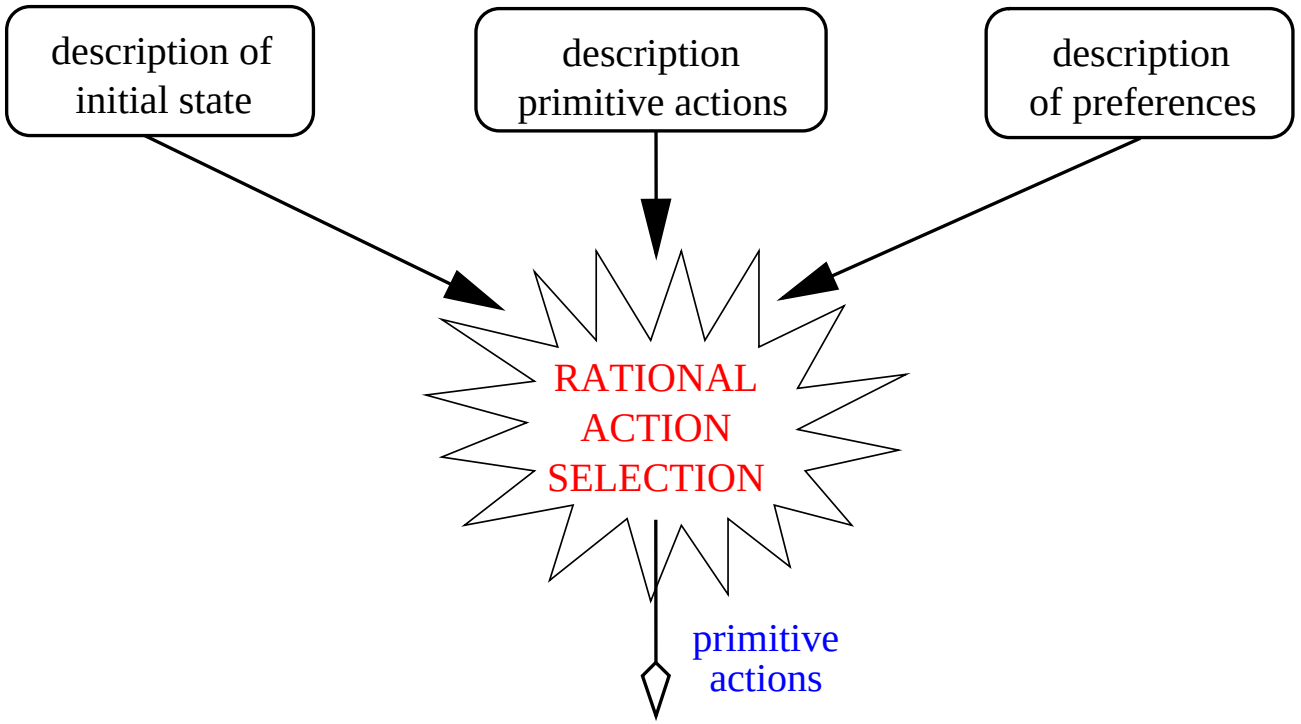
The study of the knowledge representation and reasoning problems faced by an autonomous agent in a dynamic and incompletely known world.



- knowledge of the agent vs. the designer?
- perception vs. reasoning?
- when is initial knowledge adequate to achieve a goal?

Do not want to simply engineer robot controllers that solve a class of problems or that work in a class of application domains.

Doing the right thing

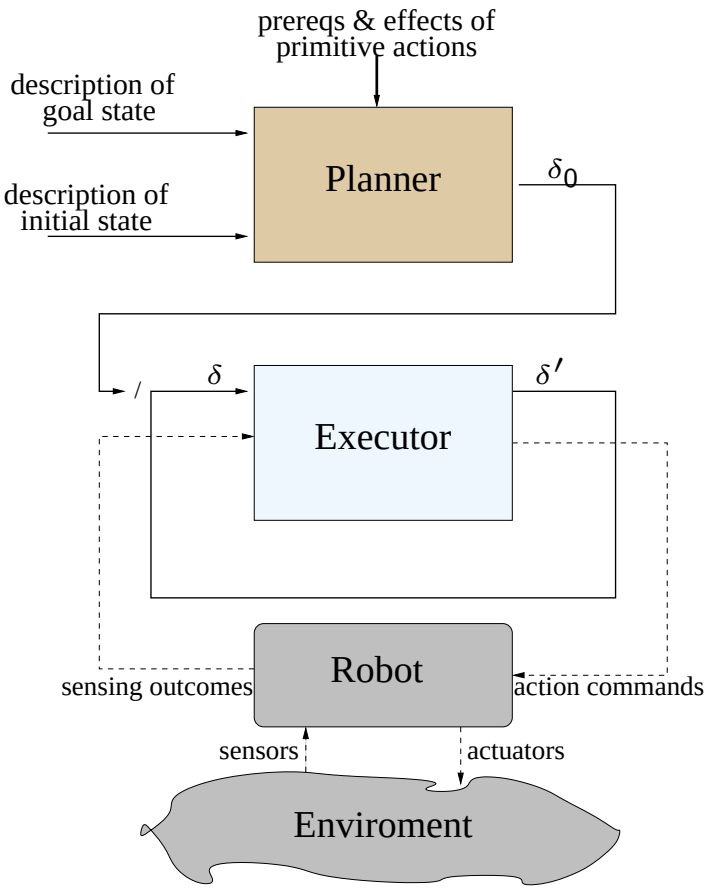


would like a system that is capable of generating actions in the world that are appropriate, as a function of some current set of beliefs and desires.

Note: the descriptions and preferences are inputs (not wired in!)

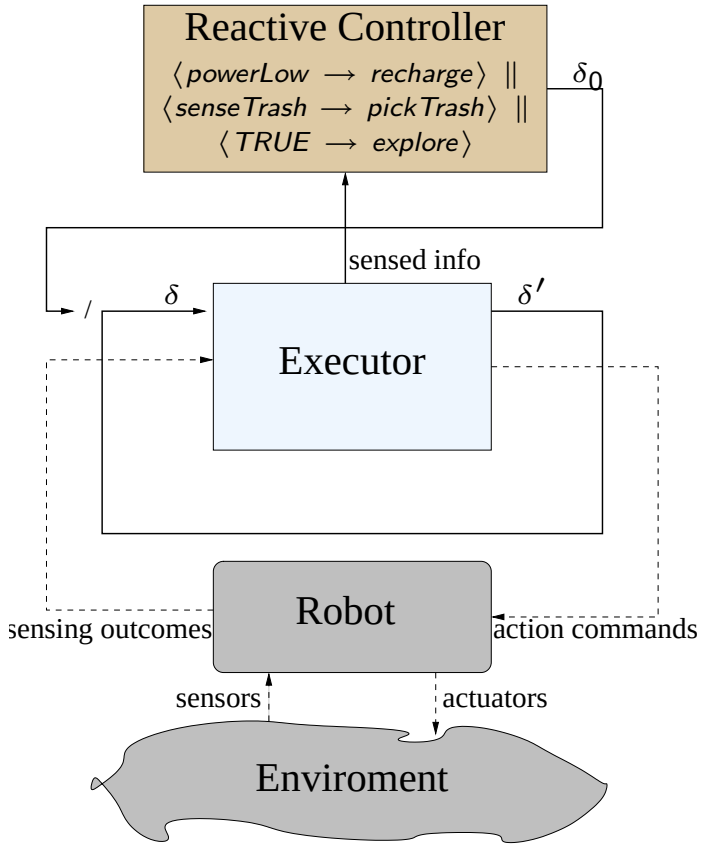
Classical Planning [SRI Shakey "The Robot" '70]

- Planing from first-principle:
  Goal oriented (usually goals-to-be)
- Dumb executor, little feedback,
  no reactivity, does not scale!
- Kind of plans δ_0:
  - Linear total-order plans (STRIPS)
  - Partial-order plans (SNLP)
  - Conditional plans (CNLP, S-GP)
  - **Robot programs** [Levesque 96]
- An interesting case: [Sacerdoti 74]
  HTN planning/programming



The Reactive Approach [Brooks 86,91]

- Reactive robots, headless!
- Little representation, quick decisions
- But: little or no pro-activity
- Examples:
  - Subsumption architecture [Brooks 86]
  - Universal plans [Schoopers 87]
  - Control Reactive Rules [Baral&Son 98]
  - (PO)MDP and policies?



AGENT = Action Theory + High-Level Program

In *Cognitive Robotics*, we think of an agent equipped with:

- An action theory: the agent knows the theory and its consequences (actions' effects, frame & qualification problems, sensing, etc.)
- A high-level program: specifying the agent tasks/behaviors (nondeterministic & domain actions)

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$$
\begin{aligned}
\mathit{getOnFlight} \stackrel{\text{def}}{=}\ & (\pi a.a)^*; \mathit{At}(\mathit{airport})?; \\
& (\mathit{go}(\mathit{terminal}1) \mid \mathit{go}(\mathit{terminal}2)); \\
& (\mathit{buy}(\mathit{magazine}) \mid \mathit{buy}(\mathit{newspaper})); \\
& \textbf{if } (\mathit{GateNo} \geq 90) \textbf{ then } \{\mathit{go}(\mathit{gate}(\mathit{GateNo})); \mathit{buy}(\mathit{coffee})\} \\
& \qquad\qquad \textbf{else } \{\mathit{buy}(\mathit{coffee}); \mathit{go}(\mathit{gate}(\mathit{GateNo}))\} \\
& \mathit{boardPlane}
\end{aligned}
$$

High-Level Programs

Programs telling the robot what needs to be done:

- primitive statements are actions to be performed by the robot:
  $\Longrightarrow$ *move to the desk and then pick up the package*
- tests within a program pertain to conditions in the world that are affected by the actions of the robot (or other agents):
  $\Longrightarrow$ **if** *the door is locked* **then** ... **otherwise** ...
  $\Longrightarrow$ **while** *there is a package on the table* **do** ...
- programs may be non-deterministic: they may contain choice points where the interpreter must make a reasoned (non-random) selection:
  $\Longrightarrow$ *go through the (appropriate) door and retrieve the package that is waiting*
  $\Longrightarrow$ *either turn left or turn right (as appropriate) ... at which point you must be located in the hall*

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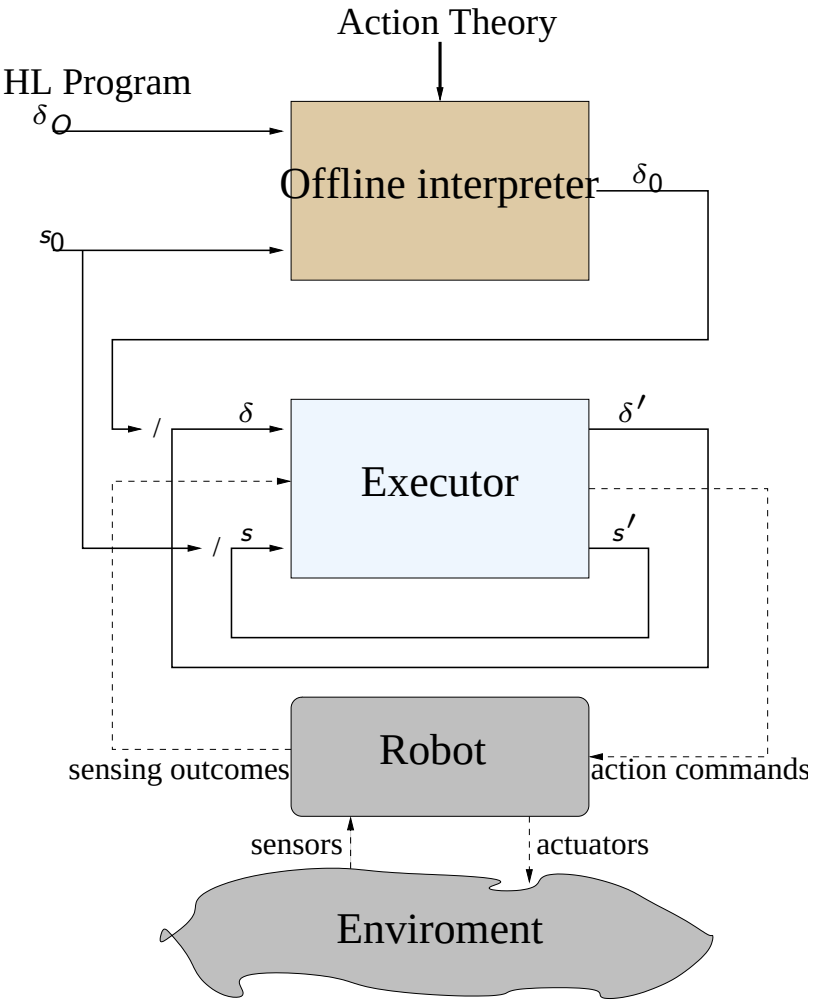
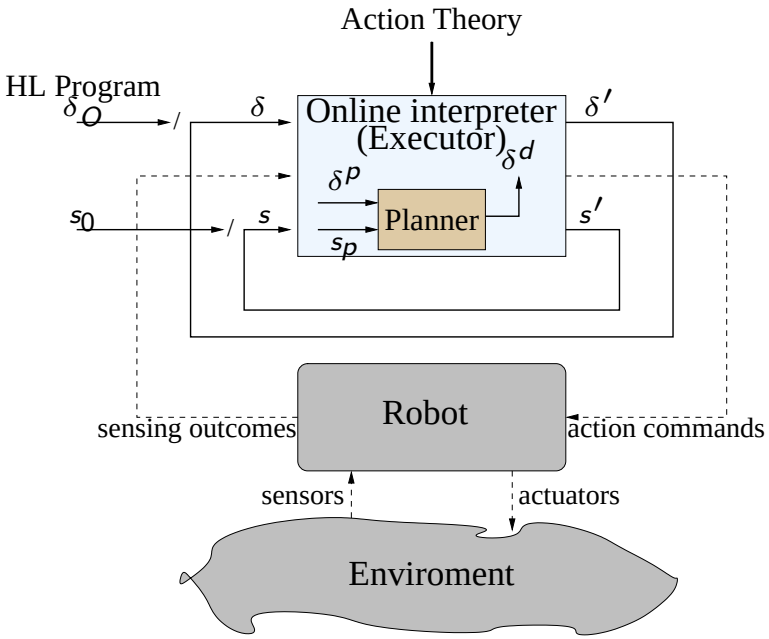
The Agent Programming Approach I

- Planning with *procedural* clues
  Goals-to-do
- Less complexity! But still hard!
- Little reactive and feedback.
- Examples:
  - GOLOG, CONGOLOG, DTGOLOG
  - MDPs + HAM [Parr 98]
  - HTN-planning [Eral et. al 94]



The Agent Programming Architecture II [INDIGOLOG]

- High-level programs executed online or incrementally
- Planning is just a *sub-module*
- Control over extended periods
- Reactive: can adapt
- Proactive: plan locally
- Can handle runtime sensing



We will talk about...

1. Action theories in the situation calculus.
2. Offline execution of high-level programs.
3. Online execution of high-level programs and local planning.

Outline

Situation Calculus [McCarthy and Hayes 69]

The description of the initial state, and the prerequisites and effects of actions can be formulated in the language of the *situation calculus*:

1. A dialect of the predicate calculus with special sorts for:
   - actions: $closeDoor$, $pickUp(package\_13)$, $moveFwd(2)$;
   - situations: S_0, the initial situation, $do(a, s)$, where a is an action term and s is a situation term.
2. Predicates or functions whose value changes from situation to situation are called fluents:

$$\neg Holding(x, \underline{S_0}) \wedge Holding(x, \underline{do(pickUp(x), S_0)})$$

3. Use a distinguished predicate $Poss(a, s)$ to assert that action a is possible to execute in situation s.

Basic Action Theories [Lin and Reiter 94]

Theory $\mathcal{D} = \Sigma \cup \mathcal{D}_{S_0} \cup \mathcal{D}_{Poss} \cup \mathcal{D}_{SSA} \cup \mathcal{D}_{UNA}$, where:

- $\mathcal{D}_{S_0}$: axioms describing the initial state of the fluents in S_0;
  $\Longrightarrow$ allows for incomplete knowledge!
- $\mathcal{D}_{poss}$: axioms describing the prerequisites of the primitive actions, of the form:
$$Poss(A(\vec{x}), s) \equiv \Theta_A(\vec{x}, s),$$
one for each primitive action A.
- $\mathcal{D}_{ssa}$: axioms describing the effects and non-effects of actions, of the form:
$$F(\vec{x}, do(a, s)) \equiv \psi_F(\vec{x}, a, s),$$
one for each fluent F.
- $\mathcal{D}_{una}$: unique name axioms for the actions.
- Σ: some foundational domain-independent axioms.

A solution to the frame problem (sometimes) [Reiter 91]

1. Positive effect axioms:

$$Holding(x, s) \wedge Fragile(x, s) \supset Broken(x, \underline{do(drop(x), s)})$$

2. Positive effect axioms:

$$HasGlue(s) \wedge Broken(x, s) \supset \neg Broken(x, \underline{do(repair(x), s)})$$

3. Frame axiom axioms:

$$\neg Broken(x, s) \wedge (y \neq x \vee \neg Fragile(x, s)) \supset \neg Broken(x, \underline{do(drop(y), s)})$$

4. Completeness assumption:

$$\neg Broken(x, s) \wedge Broken(x, do(a, s)) \supset Fragile(x, s) \wedge a = drop(x)$$

Successor state axioms = positive effect axioms + positive effect axioms + frame axioms + completness assumption

A solution to the frame problem (sometimes) [Reiter 91]

So, suppose that for fluent $F(\vec{x}, s)$:

$\gamma_F^+(\vec{x}, a, s)$ encodes all *positive* effects; and
$\gamma_F^-(\vec{x}, a, s)$ encodes all *negative* effects.

The successor state axiom solution has the following form:

$$F(\vec{x}, do(a, s)) \equiv \gamma_F^+(\vec{x}, a, s) \vee F(\vec{x}, s) \wedge \neg\gamma_F^-(\vec{x}, a, s)$$

In our example:

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In our example:

$$\gamma_{Broken}^+(\vec{x}, a, s) \stackrel{\text{def}}{=} \ldots$$
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$$Broken(x, do(a, s)) \equiv \gamma_{Broken}^+(\vec{x}, a, s) \vee Broken(\vec{x}, s) \wedge \neg\gamma_{Broken}^-(\vec{x}, a, s)$$

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$$\gamma_{Broken}^+(\vec{x}, a, s) \stackrel{\text{def}}{=} a = drop(x) \wedge Holding(x, s) \wedge Fragile(x, s)$$
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Reasoning about actions

- Projection task: *given a sequence of actions* $a_1, \ldots, a_n$, *does condition* $\phi(s)$ *holds?*:

$$\mathcal{D} \models \phi(do(do(a_n, do(a_{n-1}, \ldots, do(a_1, S_0)) \ldots)$$

- Deductive planning: *find a sequence of actions to make* $\phi(s)$ *true*:

$$\mathcal{D} \vdash \exists s.\phi(s) \wedge executable(s)$$

- Database transaction formalization:

$$\mathcal{D} \models \exists c.Enrolled(john, c, do(reg(mary, c10), do(drop(john, c21), S_0)))$$

- Inductive proofs:

$$\mathcal{D} \models \forall p, n, n'.s \sqsubseteq s' \wedge sal(p, n, s) \wedge sal(p, n', s') \supset n \leq n'$$

Reasoning about actions (cont.)

- Knowledge and beliefs: what does *the agent* knows?

$$\phi(s) \text{ vs } \mathbf{Know}(\phi, s).$$

- Explicit time and continuous change:

$$start(s) = t \text{ and } Velocity(x, s) = 87.$$

- Natural actions: actions that *must* occur (bouncing of the ball).
- Concurrent actions:

$$\mathcal{D} \models \neg Spilled(coffee, do(\{liftRight(table), liftLeft(table)\}, S_0),$$
$$\mathcal{D} \models Spilled(coffee, do(\{liftRight(table)\}, S_0).$$

- Ramifications: indirect effects

$$\mathcal{D} \models \forall s. Inside(x, y) \supset At(x, l, do(move(y, l), S_0)).$$

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All in plain classical (i.e., predicate) logic!

Outline

GOLOG [Levesque 96]

GOLOG is a *logic-based* programming language whose primitive actions are those of a background action theory.

It includes the following constructs:

α,	*primitive action*
$\phi?$,	*test/wait for a condition*
$(\delta_1; \delta_2)$,	*sequence*
if ϕ **then** δ_1 **else** δ_2 **endIf**,	*conditional*
while ϕ **do** δ **endWhile**,	*loop*
proc $\beta(\vec{x})$ δ **endProc**,	*procedure definition*
$\beta(\vec{t})$,	*procedure call*
$(\delta_1 \mid \delta_2)$,	*nondeterministic choice of action*
$\pi\, \vec{x}\, [\delta]$,	*nondet. choice of arguments*
δ^*,	*nondeterministic iteration*

GOLOG semantics

An offline execution of a GOLOG program δ relative to an action theory $\mathcal{D}$, is a situation term S such that:

$$\mathcal{D} \models Do(\delta, S_0, S).$$

$Do(\delta, s, s')$: *it is possible to reach situation* s' *from situation* s *by executing a sequence of actions specified by* δ.

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This execution formula can be defined inductively:

$$\begin{aligned}
Do(a, s, s') &\stackrel{\text{def}}{=} Poss(a, s) \wedge s' = do(a, s) \\
Do(?\phi, s, s') &\stackrel{\text{def}}{=} \phi(s) \wedge s' = s \\
Do(\delta_1; \delta_2, s, s') &\stackrel{\text{def}}{=} \exists s''. Do(\delta_1, s, s'') \wedge Do(\delta_2, s'', s') \\
Do(\delta_1 \mid \delta_2, s, s') &\stackrel{\text{def}}{=} Do(\delta_1, s, s') \vee Do(\delta_2, s, s') \\
Do(\pi x.\delta(x), s, s') &\stackrel{\text{def}}{=} \exists x. Do(\delta(x), s, s')
\end{aligned}$$

GOLOG implementation

Do can be *lifted* directly from its semantic specification:

```
do([E1,E2],S,S1) :- do(E1,S,S2), do(E2,S2,S1).
do(?(P),S,S) :- holds(P,S).
do(ndet(E1,E2),S,S1) :- do(E1,S,S1) ; do(E2,S,S1).
do(if(P,E1,E2),S,S1) :- do(ndet([?(P),E1],[?(-P),E2]),S,S1).
do(star(E),S,S1) :- S1 = S ; do([E,star(E)],S,S1).
do(while(P,E),S,S1):- do([star([?(P),E]),?(-P)],S,S1).
do(pi(V,E),S,S1) :- sub(V,_,E,E1), do(E1,S,S1).
do(E,S,S1) :- proc(E,E1), do(E1,S,S1).
do(E,S,do(E,S)) :- prim_action(E), poss(E,P), holds(P,S)

holds(and(P,Q),S) :- holds(P,S), holds(Q,S).
...
```

Some GOLOG applications

A number of applications have been written in GOLOG and variants:

- a simulated elevator controller;
- mail delivery robots at Toronto and York Universities
  $\Longrightarrow$ demonstrates hardware independence
- a robot museum guide [Burgard et al AAAI 98]
  $\Longrightarrow$ controllable online via the Web
- business process simulation [Yu & Mylopoulos 97]
  $\Longrightarrow$ for analysis of the processes, not execution
- characters in computer animation [Funge SIGGRAPH 99]
  $\Longrightarrow$ for high-level behaviour specification
- a large "softbot" application [Ruman MSc 96]
  personal banking: ATM controller, personal agents, bank agents, routers agents run as processes under Unix, communicate over TCP/IP

Why is Golog popular?

[Copied from Ryan Kelly's presentation, 2005]

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- Logic-based:
  - Compact implementation in Prolog.
  - Possible to prove safety/liveness properties.

Outline

What is missing?

A number of concerns are not well addressed by these efforts:

1. sensing and exogenous actions
2. reactivity: suspending some preprogrammed behaviour
   $\Longrightarrow$ e.g., monitoring power consumption
3. reasoned failure recovery
   $\Longrightarrow$ e.g., encountering an unexpected closed door
4. reliable and non-inmediate primitive actions
   $\Longrightarrow$ e.g., $go\_to(loc)$ **vs.** $start\_going\_to(loc)$ + exog. reports
5. local decision making
   $\Longrightarrow$ e.g., searching through an entire execution **vs.** committing quickly to the first action

These issues are addressed in more recent dialects of GOLOG.

Reactivity in CONGOLOG [de Giacomo et al AIJ 00]

An extension to GOLOG, which includes concurrency, prioritized interrupts, and exogenous actions.

$$\begin{array}{rcl}
\langle\ \mathit{TooHot} \wedge \neg \mathit{FanOn} & \rightarrow & \mathit{toggle_fan}\ \rangle\rangle \\
\langle\ \mathit{TooCold} \wedge \mathit{FanOn} & \rightarrow & \mathit{toggle_fan}\ \rangle\ \rangle \\
\langle\ \mathit{SmokeDetectorOn} & \rightarrow & \mathit{ring_alarm}\ \rangle\ \rangle \\
\langle\ n\ :\ \mathit{ButtonOn}(n) & \rightarrow & \pi f.?(\mathit{BestButton}(f)); \mathit{serve}(f)\rangle\ \rangle \\
\langle\ \neg \mathit{OnFloor}(1) & \rightarrow & \mathit{down}\rangle
\end{array}$$

Allows reactive controllers, but *still via reasoning about actions*!

- $\delta_1 \| \delta_2$: concurrent execution (equal priority)
- $\delta_1 \rangle\rangle \delta_2$: concurrent execution ($\delta_1$ with higher priority)
- $\delta^{\|}$: concurrent iteration
- $\langle\ \vec{x} : \phi(\vec{x}) \rightarrow \delta(\vec{x})\ \rangle$: interrupt

Semantics of CONGOLOG: *Trans* and *Final*

To characterize CONGOLOG behaviour, we need a theory of complex actions based on single steps:

1. $Trans(\delta, s, \delta', s')$: program δ can legally execute a single step to s', with δ' remaining to execute.
2. $Final(\delta, s)$: program δ can legally terminate.

$$Trans(a, s, \delta', s') \equiv Poss(a, s) \wedge s' = s \wedge \delta' = nil$$

$$Trans(\delta_1 \| \delta_2, s, \delta', s') \equiv \exists \delta''. Trans(\delta_1, s, \delta'', s') \wedge \delta' = (\delta''; \delta_2) \vee Trans(\delta_2, s, \delta'', s') \wedge \delta' = (\delta_1; \delta'')$$

$$Final(\textbf{while } \phi \textbf{ do } \delta, s) \equiv \neg\phi(s)$$

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$$\mathcal{D} \cup \mathcal{C} \models \exists s'. Do(\delta, s, s'),$$

where $Do(\delta, s, s') \stackrel{\text{def}}{=} \exists \delta'. Trans^*(\delta, s, \delta', s') \wedge Final(\delta', s')$

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$$Trans(\delta_1 \| \delta_2, s, \delta', s') \equiv \exists \delta''. Trans(\delta_1, s, \delta'', s') \wedge \delta' = (\delta''; \delta_2) \vee Trans(\delta_2, s, \delta'', s') \wedge \delta' = (\delta_1; \delta'')$$

$$Final(\textbf{while } \phi \textbf{ do } \delta, s) \equiv \neg\phi(s)$$

$$Final(\delta^*, s) \equiv TRUE$$

$\mathcal{D} \cup \mathcal{C} \models \exists s'. Do(\delta, s, s')$, ***** STILL OFFLINE! *****

where $Do(\delta, s, s') \stackrel{\text{def}}{=} \exists \delta'. Trans^*(\delta, s, \delta', s') \wedge Final(\delta', s')$

Outline

INDIGOLOG [Sardina 2000,2004]

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We cannot compute an entire execution of a program offline before starting. (This was impractical for large programs anyway!)

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Top level interpreter for INDIGOLOG:

```
indigo(P,S):- exog_occurs(A), !, indigo(P,[A|S]).
indigo(P,S):- final(P,S), !.
indigo(P,S):- trans(P,S,P1,S), !, indigo(P1,S).
indigo(P,S):- trans(P,S,P1,[A|S]),exec(A,R),indigo(P1,[A|S])
```

Local search $\Sigma(\delta)$

Unlike ordinary GOLOG, INDIGOLOG handles non-determinism by making a random selection:

> *The non-deterministic program* $(\delta_1 \mid \delta_2); \ldots; p?$ *may fail if the wrong action is selected and executed!*

The solution: extend the language with a new operator Σ where $\Sigma(\delta)$ means "take transitions in δ that will ultimately succeed":

$$Trans(\Sigma(\delta), s, \delta', s') \equiv Trans(\delta, s, \delta', s') \wedge (\exists s'')Do(\delta', s', s'').$$

Can control the amount of lookahead search:

$$\Sigma(\delta_1); \delta_2 \quad \textbf{vs} \quad \Sigma(\delta_1; \delta_2)$$

An example of local planning

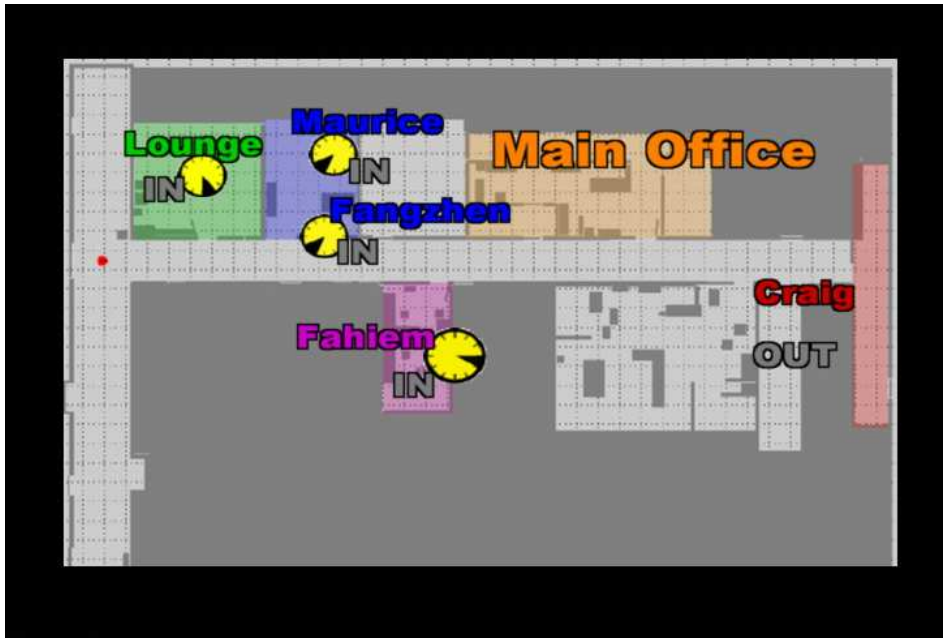
$$getOnFlight'_{Sketchy} \stackrel{\text{def}}{=}$$

$$\begin{array}{l}
\Sigma[(\pi a.a)^*; At(airport)?]; \\
\Sigma[(go(term1) \mid go(term2)); \\
\quad checkDepartures^*; \qquad /^* \text{ Sensing Action! } ^*/ \\
\quad (buy(magazine) \mid buy(newspaper)); \\
\quad \textbf{if } GateNo \geq 90 \textbf{ then } \{go(gate(GateNo)); buy(coffee)\} \\
\qquad\qquad\qquad\qquad \textbf{else } \{buy(coffee); go(gate(GateNo))\} \ ]; \\
boardPlane
\end{array}$$

Other work in the area

1. Execution monitoring: want to compare the results of executing GOLOG program to what was expected [de Giacomo et al 98]
2. Limited planning: would like to be able to introduce limited forms of planning as part of much larger GOLOG programs.
   use ideas of [Bacchus & Kabanza 96] for filtering
3. Knowledge/belief and knowledge-base programming:
   [Moore 85, Scherl & Levesque 92, Shapiro et al 00, Reiter 01]
4. Decision-theoretic GOLOG: DTGOLOG [Boutilier et al AAAI 2000]
5. Plan correctness: physical+epistemic feasibility [Sardina et al 04,06]

Golem running GOLOG



Golem

Outline

Conclusions

Up until very recently, "robot programming" was tackled mainly by engineers and hobbyists:

- required familiarity with workings of sensors and effectors, and
- programming languages very close to hardware.

(not unlike situation with first computers.)

Increasingly, because of:

- more stable robotic platforms, and
- better low-level interfaces and simulations,

it has become possible to consider robot programs that are much more portable and hardware independent.

Conclusions (cont.)

Situation calculus provides:

1. an expressive framework to reason about action and change;
2. several high-level agent programing languages.

Plus:

- √ everything with a clean semantics in classical logic!
- √ elaboration tolerant (many extensions in a parsimonious way).

Robotics by computer scientists!:

- abstraction, encapsulation, modularity;
- semantics / correctness, computability.